Year 10 Higher

# REVISION PACK

## Graphs 2

**Name:** _____

**Class:** _____

## Graphs 2 · Distance-time graphs (Lesson 8.2)

**Worked example: time stopped and speed home**

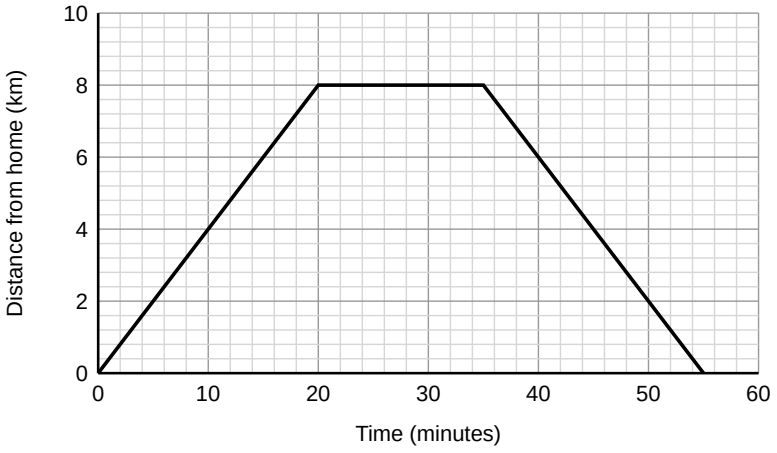
A cyclist rides 9 km from home in 30 minutes, stops from 30 to 45 minutes, then rides home in 30 minutes.

(a) How long did they stop? (b) Work out the average speed on the way home in km/h.

| | | | |
|---|---|---|---|
| (a) | Step 1 | Flat line = stopped | $45 - 30 = 15$ |
| | Step 2 | Answer | $15$ minutes |
| (b) | Step 1 | Time in hours | $30 \text{ min} = \frac{1}{2} \text{ h}$ |
| | Step 2 | Distance ÷ time | $9 \div \frac{1}{2} = 18$ |
| | Step 3 | Answer | $18 \text{ km/h}$ |

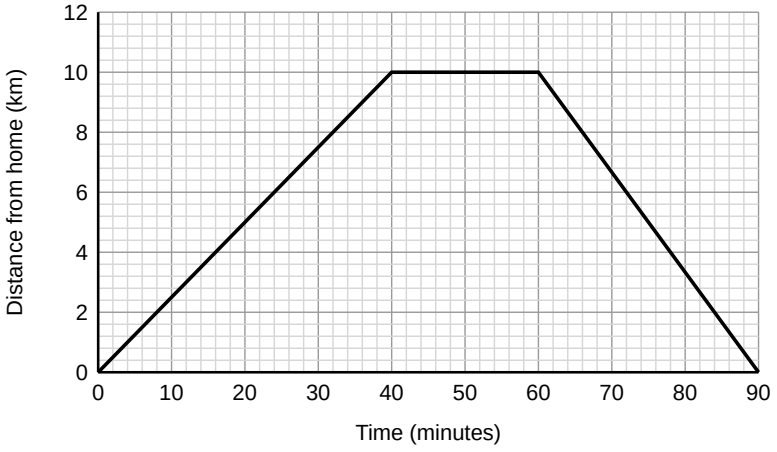
### Practice: time stopped and speed home

**1.1**




Ali cycles from home to a shop and back.
The distance-time graph shows the journey.
(a) How long did Ali stay at a shop?
(b) Work out Ali's average speed on the way home.
Give your answer in km/h.

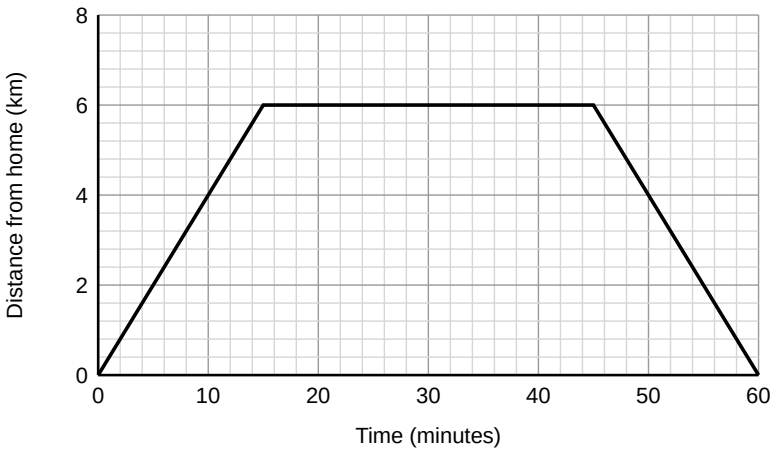
**1.2**




Sara cycles from home to a friend's house and back.
The distance-time graph shows the journey.
(a) How long did Sara stay at a friend's house?
(b) Work out Sara's average speed on the way home.
Give your answer in km/h.

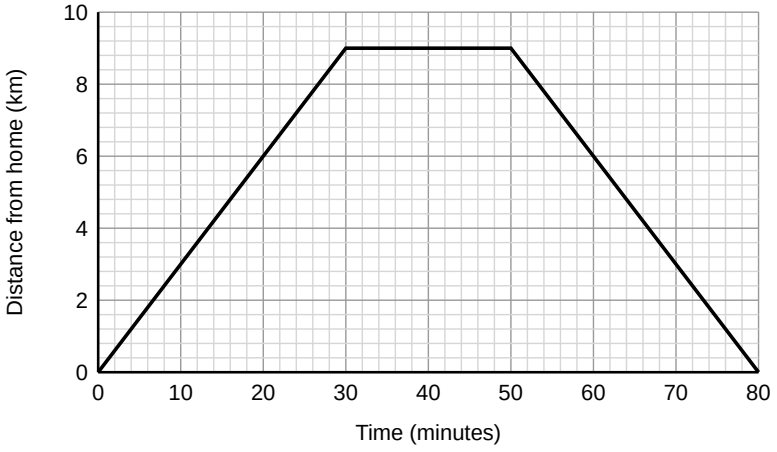
## Practice: time stopped and speed home (continued)

**1.3**




Omar cycles from home to the beach and back.
The distance-time graph shows the journey.
(a) How long did Omar stay at the beach?
(b) Work out Omar's average speed on the way home.
Give your answer in km/h.

**1.4**




Huda cycles from home to the library and back.
The distance-time graph shows the journey.
(a) How long did Huda stay at the library?
(b) Work out Huda's average speed on the way home.
Give your answer in km/h.

**Worked example: acceleration and total distance**

A car speeds up from 0 to 10 m/s in 4 s, travels at 10 m/s for 6 s, then stops in 2 s.

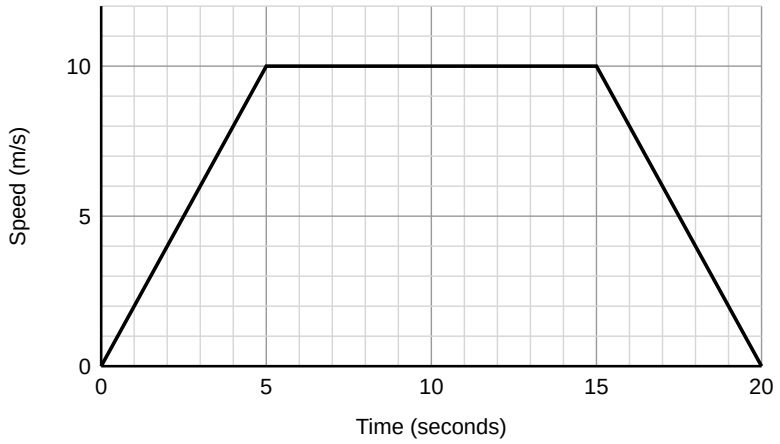
(a) Work out the acceleration in the first 4 seconds.

(b) Work out the total distance travelled.

| | | | |
|---|---|---|---|
| **(a)** | **Step 1** | Speed ÷ time | $10 \div 4 = 2.5$ |
| | **Step 2** | Answer | $2.5\ \text{m/s}^2$ |
| **(b)** | **Step 1** | Triangle | $\frac{1}{2} \times 4 \times 10 = 20$ |
| | **Step 2** | Rectangle | $6 \times 10 = 60$ |
| | **Step 3** | Triangle | $\frac{1}{2} \times 2 \times 10 = 10$ |
| | **Step 4** | Add | $20 + 60 + 10 = 90$ |
| | **Step 5** | Answer | $90\ \text{m}$ |

## Practice: acceleration and total distance

**2.1**

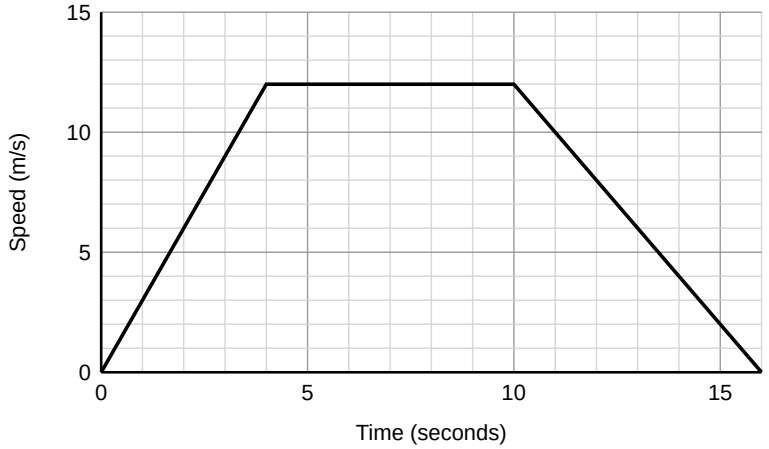



The speed-time graph shows a car journey of 20 seconds.

(a) Work out the acceleration of the car during the first 5 seconds.

(b) Work out the total distance travelled by the car.

**2.2**



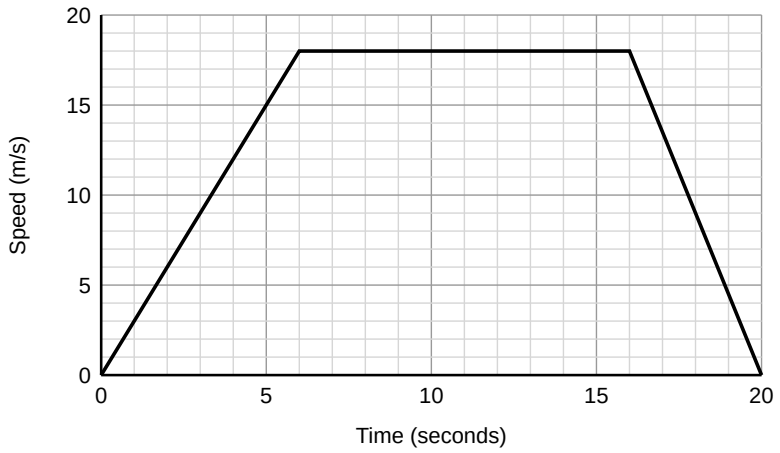

The speed-time graph shows a car journey of 16 seconds.

(a) Work out the acceleration of the car during the first 4 seconds.

(b) Work out the total distance travelled by the car.

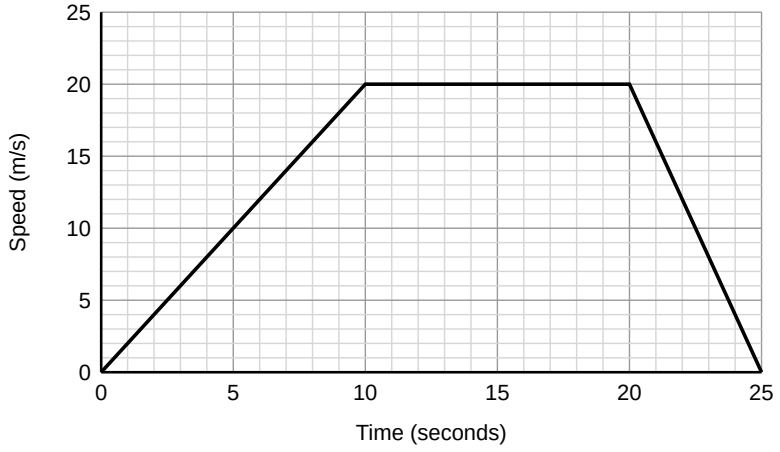
## Practice: acceleration and total distance (continued)

**2.3**




The speed-time graph shows a car journey of 20 seconds.
(a) Work out the acceleration of the car during the first 6 seconds.
(b) Work out the total distance travelled by the car.

**2.4**




The speed-time graph shows a car journey of 25 seconds.
(a) Work out the acceleration of the car during the first 10 seconds.
(b) Work out the total distance travelled by the car.

## Graphs 2 · Perpendicular lines (Lesson 8.4)

**Worked example: equation of a perpendicular line**

The line $L$ has equation $y = 3x - 1$. Find an equation of the line that is perpendicular to $L$ and passes through the point (6, 4).

| | | |
|---|---|---|
| **Step 1** | Gradient of $L$ | $3$ |
| **Step 2** | Flip and change sign | $-\frac{1}{3}$ |
| **Step 3** | Put in (6, 4) | $4 = -\frac{1}{3} \times 6 + c$ |
| **Step 4** | Find $c$ | $c = 4 + 2 = 6$ |
| **Step 5** | Answer | $y = -\frac{1}{3}x + 6$ |

### Practice: equation of a perpendicular line

**3.1** The line $L$ has equation $y = 2x + 5$. Find an equation of the line that is perpendicular to $L$ and passes through the point (6, 1).

**3.2** The line $L$ has equation $y = 4x - 3$. Find an equation of the line that is perpendicular to $L$ and passes through the point (8, 5).

**3.3** The line $L$ has equation $y = -3x + 2$. Find an equation of the line that is perpendicular to $L$ and passes through the point (3, 4).

**3.4** The line $L$ has equation $y = 5x + 1$. Find an equation of the line that is perpendicular to $L$ and passes through the point (10, 3).

**3.5** The line $L$ has equation $y = -2x + 4$. Find an equation of the line that is perpendicular to $L$ and passes through the point (4, 7).

**3.6** The line $L$ has equation $y = 3x - 2$. Find an equation of the line that is perpendicular to $L$ and passes through the point (6, 0).

## Graphs 2 · Mid-points and length of a line (Lessons 8.5, 8.6)

**Worked example: mid-point and length**

$A$ is the point (1, 2). $B$ is the point (7, 10).

(a) Find the coordinates of the midpoint of $AB$. (b) Work out the length of $AB$.

| | | | |
|---|---|---|---|
| **(a)** | **Step 1** | Add $x$, halve | $\frac{1+7}{2} = 4$ |
| | **Step 2** | Add $y$, halve | $\frac{2+10}{2} = 6$ |
| | **Step 3** | Answer | $(4, 6)$ |
| **(b)** | **Step 1** | Across and up | $7 - 1 = 6, \quad 10 - 2 = 8$ |
| | **Step 2** | Pythagoras | $\sqrt{6^2 + 8^2} = \sqrt{100}$ |
| | **Step 3** | Answer | $10$ |

### Practice: mid-point and length

**4.1** $A$ is the point (−1, 2). $B$ is the point (5, 10).

(a) Find the coordinates of the midpoint of $AB$. (b) Work out the length of $AB$.

**4.2** $A$ is the point (−3, 4). $B$ is the point (9, −1).

(a) Find the coordinates of the midpoint of $AB$. (b) Work out the length of $AB$.

**4.3** $A$ is the point (2, −3). $B$ is the point (6, 4).

(a) Find the coordinates of the midpoint of $AB$. (b) Work out the length of $AB$. Give your answer correct to 1 decimal place.

**4.4** $A$ is the point (−1, −1). $B$ is the point (2, 3).

(a) Find the coordinates of the midpoint of $AB$. (b) Work out the length of $AB$.

**4.5** $A$ is the point (0, −4). $B$ is the point (5, 8).

(a) Find the coordinates of the midpoint of $AB$. (b) Work out the length of $AB$.

**4.6** $A$ is the point (−6, 1). $B$ is the point (2, 7).

(a) Find the coordinates of the midpoint of $AB$. (b) Work out the length of $AB$.